Luca **Bacco**

AI Research Scientist

Rome, Italy

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Professional Summary

- **AI Research Scientist** with 7+ years of experience designing, implementing, and evaluating machine and deep learning models, including LLMs, mainly for NLP, multimodality, and explainable AI, with main focus on biomedical applications.
- Proven track record of **40+ peer-reviewed** publications, **70%** as lead-author in **Q1 journals** and international conferences.
- End-to-end experience on **€1M-funded research**, from proposal to **multidisciplinary** research coordination.
- Proven record of **improving** model performance in low-data and specialized-domain settings, with focus on biomedicine.
- Experienced in **rapidly adapting** research methods across domains and **mentoring** multidisciplinary PhD and thesis research, with supervised work repeatedly leading to peer-reviewed publications.
- Strong hands-on expertise in Python, Hugging Face, PyTorch, TensorFlow.

Google Scholar: h-index: 12 | 432 citations | user: Aw4cEzMAAAAJ

Work Experience

Università Campus Bio-Medico di Roma, Department of Engineering

Assistant Professor — Sep. 2025 – present

- Lead multidisciplinary AI research on LLMs, NLP, multimodal learning, computer vision, physiological signals, and generative AI.
- Supervise 1 PhD student and mentor 6 more, contributing to 10 accepted conference papers.
- Supervised 5 thesis projects, turning 2 of them into peer-reviewed publications and 2 manuscripts in preparation.
- Serve as Internationalization Coordinator and in the Teaching Quality Control Unit for the Biomedical Engineering MSc, aligning faculty and institutional stakeholders.

Postdoctoral Researcher — Nov. 2022 – Aug. 2025

- Awarded 2 competitive research fellowships and 1 scholarship.
- Published 15 peer-reviewed works (13 journal articles) on multimodal AI, LLMs, explainable AI, and time-series modeling.
- Supervised/co-supervised 11 student theses and mentored 2 research fellows, both progressed to PhD positions.

PhD Candidate — Nov. 2019 – Nov. 2022

- Conducted research in explainable AI, with main focus on NLP and Transformers, resulting in 15 peer-reviewed publications (9 journal).
- Co-supervised 5 student theses, turning 60% of them into peer-reviewed publications.

University of Groningen, Computational Linguistics Department

Visiting PhD Student — Jan. 2022 – July 2022

- Conducted NLP research on Transformer reliability, domain adaptation, and text style transfer under Prof. Malvina Nissim, achieving 2 first-author Q1 journal publications and initiating international collaborations with researchers and PhD students.
- Completed advanced NLP training and weekly research seminars; guided a student team project awarded as the Best Student Project.

Education

Università Campus Bio-Medico di Roma

PhD in Information Technology for Biomedicine — Mar. 2023

Thesis: *Exploring New Technologies in Healthcare: Advancing Natural Language Processing* – Investigating Transformer-based NLP for biomedical applications with emphasis on explainability, domain adaptation, and medical text simplification.
Supervisors: Felice Dell'Orletta (Istituto di Linguistica Computazionale, CNR), Mario Merone (Università Campus Bio-Medico di Roma).

MSc – Biomedical Engineering (110/110 cum laude) — Feb. 2019

Thesis: *Development of a motion control system for an anthropomorphic manipulator based on online extraction of muscle synergies*

Key Projects

[XAI-CARE] AI Researcher in Biomedicine — Jan. 2023 – Feb. 2026

- Contributed across the full lifecycle of the €1M-funded project (proposal to final reporting).
- Guided 2 research fellows in developing clinical NLP, computer vision, and multimodal AI research, coordinating with multidisciplinary stakeholders, resulting in 4 published papers, 1 manuscript currently under review, and 1 aligned multimodal Italian clinical dataset.

[Digital Innovation in Price Statistics] AI Researcher in Economics — Mar. 2024 – Aug. 2025

- Coordinated university-industry research on data pipelines and time-series modeling, resulting in 3 journal papers, 2 manuscripts under review, together with 1 publicly released semi-automatically labeled dataset and 1 synthetic dataset for supporting additional research.

[DRIVERS] **Full-stack Developer** Nov. 2022 – Mar. 2024

- Built a web platform from scratch, implemented backend and frontend components, application database, external API integrations, and an integrated web-scraping and data-ingestion pipeline.

[DHelp4H] **NLP Early-stage Researcher** Nov. 2019 – Nov. 2022

- Coordinated university-industry R&D to integrate NLP into a company product; dissemination to regional public-sector stakeholders.
- Designed and developed 2 clinical use cases for an NLP-based FAQ-answering chatbot.

Teaching Experience

Lecturer & Teaching Assistant 2020 – present

- Taught across 20 BSc, MSc, and postgraduate courses in ML, DL, NLP, Generative AI, biomedical data processing, and programming.
- Delivered 30+ hours of professional and outreach teaching across 4 activities for academic, industry, and public-sector audiences.
- Supervised/co-supervised 20+ thesis projects, turning 50%+ of them into peer-reviewed publications and ongoing research works.

Skills

Languages:	Italian (native); English (professional proficiency)
Machine Learning:	Large Language Models (LLMs); Transformers; Fine-tuning; Representation Learning; Distillation; Retrieval-Augmented Generation (RAG)
Domains:	NLP; Computer Vision; Time-series modeling; Multimodality; Biomedical AI; Explainability
Programming:	Python (Hugging Face, PyTorch, TensorFlow, Keras, scikit-learn, NumPy); C#; C/C++; JavaScript; SQL
Engineering:	ASP.NET; MATLAB and Simulink; MongoDB; LaTeX; Linux; GitHub; Bash; web scraping; HTML; CSS;
Soft Skills:	Critical Thinking; Interdisciplinary communication; Mentoring; Continuous learning; Analytical mindset; Versatility; Public speaking; Teamwork; Academic writing

Selected Publications

Cross-Lingual Distillation for Domain Knowledge Transfer with Sentence Transformers Knowledge-Based Systems
(Co-first author, 2025) Designed a distillation framework for low-data biomedical NLP, improving F1 by 10+ points with 10% of tuning data.

From Pre-training to Fine-tuning: An In-depth Analysis of Large Language Models in the Biomedical Domain Artificial Intelligence in Medicine
(Co-first author, 2024) Benchmarked encoder- and decoder-based LLMs with probing-based explainability, showing higher data efficiency for encoders and stronger gains from domain adaptation on biomedical tasks

A Text Style Transfer System for Reducing the Physician-Patient Expertise Gap Expert Systems with Applications
(First author, 2023) Built a representation-learning pipeline to mine pseudo-parallel expert-lay medical data and fine-tune BART, outperforming SOTA by +38 pp in lay-rated simplification and by +17 pp on physician-rated content preservation.

On the Instability of Further Pre-Training: Does a Single Sentence Matter to BERT? Natural Language Processing Journal
(First author, 2023) Demonstrated that one-step, single-sentence pre-training can alter BERT fine-tuning; performance effects diminish with more tuning data, while model divergence and training trajectories remain distinct.

Research Impact

Patent Application n. 2502168.4 13 Feb. 2025

Co-inventor of “Artificial intelligence system for non-invasive detection of calcium levels via ECG”.

Additional Experience

BPCOmedia S.r.l.

Startup Ambassador Oct. 2019 – Nov. 2019

- Participated in 1:1 mentoring sessions with investors. Delivered 2 investor-facing pitches and generated 3 partnership opportunities.

Full-Stack Developer May 2019 – Sep. 2019

- Built end-to-end full-stack web features for a web platform, including APIs integration, mobile connectivity, and responsive views.
- Reverse engineered a third-party application and produced technical documentation supporting maintenance and future development.

Interests

Naturally curious and drawn to new perspectives, I enjoy exploring philosophy, psychology, and sociology, reading, solving crosswords, and critically engaging with visual art. I also occasionally play chess, a long-standing interest since winning my high-school tournament. These interests broaden my perspective and sharpen my critical thinking.

Rome, 14th August 2026 **Signature**